

Chapter 5 Test model

Multiple Choice Questions (70)

1. In the relational model, what is the formal term for a "row" in a table?
 - a. Attribute
 - b. Domain
 - c. Tuple
 - d. Schema

2. Which of the following best defines a relation schema $R(A_1, A_2, \dots, A_n)$?
 - a. A specific set of tuples satisfying integrity constraints
 - b. The name of the relation and its set of attributes
 - c. The Cartesian product of all attribute domains
 - d. A snapshot of the database at a given time

3. What does the notation $r(R) \subset \text{dom}(A_1) \times \text{dom}(A_2) \times \dots \times \text{dom}(A_n)$ represent?
 - a. The schema of relation R
 - b. The domain of attribute A_1
 - c. The state (extension) of relation R
 - d. The superkey of relation R

4. Which of the following is NOT a characteristic of relations in the relational model?
 - a. Tuples are unordered
 - b. Attribute values must be atomic
 - c. Duplicate tuples are allowed
 - d. Attributes are ordered in the formal definition

5. If $\text{dom}(A_1) = \{0,1\}$ and $\text{dom}(A_2) = \{x,y\}$, how many elements are in $\text{dom}(A_1) \times \text{dom}(A_2)$?
 - a. 2
 - b. 3
 - c. 4
 - d. 6

-
6. What is the role of a null value in a relational tuple?
- a. It represents a default value
 - b. It indicates an unknown, inapplicable, or missing value
 - c. It enforces referential integrity
 - d. It is used only in primary keys
-

7. How is the value of attribute A_i in tuple t denoted?
- a. $t\{A_i\}$
 - b. $t[A_i]$ or $t.A_i$
 - c. $R(A_i)$
 - d. $\text{dom}(t, A_i)$
-

8. Which constraint ensures that no two tuples in a relation have the same value for a set of attributes SK?
- a. Domain constraint
 - b. Entity integrity constraint
 - c. Superkey constraint
 - d. Referential integrity constraint
-

9. A minimal superkey is also known as:
- a. Candidate key
 - b. Foreign key
 - c. Composite key
 - d. Surrogate key
-

10. Consider relation $\text{CAR}(\text{State}, \text{Reg\#}, \text{SerialNo}, \text{Make}, \text{Model}, \text{Year})$. If both $\{\text{SerialNo}\}$ and $\{\text{State}, \text{Reg\#}\}$ uniquely identify tuples, which statement is true?
- a. Only $\{\text{SerialNo}\}$ is a superkey
 - b. $\{\text{SerialNo}, \text{Make}\}$ is a candidate key
 - c. Both are candidate keys
 - d. Neither can be a primary key
-

11. What is the general rule for choosing a primary key among candidate keys?

- a. Choose the one with the most attributes
 - b. Choose the largest in terms of storage size
 - c. Choose the smallest in terms of size
 - d. Choose the one containing nulls
-

12. In a relational database schema $S = \{R_1, R_2, \dots, R_n\}$, what does S represent?

- a. A single relation state
 - b. The set of all tuples in the database
 - c. The name of the entire database schema
 - d. The set of all foreign keys
-

13. A relational database state $DB = \{r_1, r_2, \dots, r_m\}$ is valid only if:

- a. All relations have the same number of attributes
 - b. Each r_i satisfies the integrity constraints in IC
 - c. All primary keys are numeric
 - d. No nulls exist in any tuple
-

14. Which integrity constraint prohibits null values in primary key attributes?

- a. Domain constraint
 - b. Referential integrity
 - c. Entity integrity
 - d. Semantic constraint
-

15. Referential integrity involves:

- a. Only one relation
 - b. Two relations: referencing and referenced
 - c. Three or more relations
 - d. Only attributes within the same tuple
-

16. In referential integrity, a foreign key value in the referencing relation must be:

- a. Always non-null
- b. Either a valid primary key in the referenced relation or null (if not part of its own PK)
- c. Always equal to a surrogate key
- d. Unique across all relations

17. If a foreign key is part of the primary key of the referencing relation, which of the following is true?

- a. It can be null
 - b. It cannot be null
 - c. It must reference a composite key
 - d. It must be a surrogate key
-

18. In a relational schema diagram, how is a referential integrity constraint typically displayed?

- a. Double underline on the foreign key
 - b. Dashed rectangle around the relation
 - c. Directed arc from FK to referenced PK
 - d. Bold attribute name
-

19. Which of the following is an example of a semantic (application-based) constraint?

- a. Primary key must be unique
 - b. Foreign key must reference an existing PK
 - c. An employee cannot work more than 56 hours per week on all projects
 - d. Attribute values must be atomic
-

20. Which SQL construct can be used to enforce semantic constraints like the 56-hour rule?

- a. CREATE TABLE
 - b. CREATE INDEX
 - c. CREATE ASSERTION
 - d. CREATE VIEW
-

21. Which update operation can violate domain, key, entity, and referential integrity constraints?

- a. DELETE
 - b. MODIFY
 - c. INSERT
 - d. All of the above
-

22. Which constraint is **only** violated by a DELETE operation?

- a. Domain constraint
 - b. Key constraint
 - c. Entity integrity
 - d. Referential integrity
-

23. When deleting a tuple whose primary key is referenced by other tuples, which action sets the referencing foreign keys to NULL?

- a. RESTRICT
 - b. CASCADE
 - c. SET NULL
 - d. REJECT
-

24. Updating a primary key is analogous to:

- a. Two INSERTs
 - b. A DELETE followed by an INSERT
 - c. A MODIFY with CASCADE
 - d. A domain check only
-

25. Which of the following is **not** an explicit schema-based constraint in the relational model?

- a. Key constraint
 - b. Entity integrity
 - c. Referential integrity
 - d. Inherent atomicity of values
-

26. What type of constraint is "relational model does not allow lists as attribute values"?

- a. Schema-based
 - b. Semantic
 - c. Inherent/implicit
 - d. Referential
-

27. In the tuple $t = \langle \text{"John"}, 123, \text{"CA"} \rangle$, if the schema is STUDENT(Name, ID, State), what is $t[\text{State}]$?

- a. "John"
- b. 123

- c. "CA"
 - d. <"CA">
-

28. The domain of an attribute defines:

- a. The relation name
 - b. The set of valid values and their format
 - c. The foreign key references
 - d. The number of tuples
-

29. Which of the following is a correct representation of a 3-tuple in relation EMP(EID, Name, Salary)?

- a. (101, "Ali", 5000)
 - b. <101, "Ali", 5000>
 - c. {101, "Ali", 5000}
 - d. [101, "Ali", 5000]
-

30. If a relation has attributes A, B, C, and {A, B} is a candidate key, which of the following must be false?

- a. {A, B, C} is a superkey
 - b. {A} is a superkey
 - c. {B, C} may or may not be a superkey
 - d. {A, B} is minimal
-

31. Which of the following best describes a surrogate key?

- a. A natural identifier like SSN
 - b. A system-assigned artificial key
 - c. A foreign key from another relation
 - d. A composite key with nulls
-

32. In the COMPANY database schema, if DEPARTMENT has a foreign key Mgr_ssn referencing EMPLOYEE.Ssn, what does this imply?

- a. Every department must have a manager who is an employee
- b. Every employee manages a department
- c. Ssn is not a primary key in EMPLOYEE
- d. Mgr_ssn cannot be null

-
33. What is the consequence if a database state violates any integrity constraint in IC?
- a. It becomes a new schema
 - b. It is considered invalid
 - c. It triggers automatic normalization
 - d. It is accepted with warnings
-

34. Which of the following is **not** stored in the relational schema?
- a. Attribute names
 - b. Primary key designation
 - c. Actual tuple data
 - d. Foreign key references
-

35. The relational model was first proposed by:
- a. Charles Bachman
 - b. E.F. Codd
 - c. Ramez Elmasri
 - d. Peter Chen
-

36. Which paper by E.F. Codd introduced the relational model?
- a. "The Entity-Relationship Model"
 - b. "A Relational Model for Large Shared Data Banks"
 - c. "Data Abstraction in DBMS"
 - d. "Foundations of Relational Theory"
-

37. In a relation, the ordering of tuples is considered:
- a. Essential for query processing
 - b. Not significant
 - c. Required for integrity
 - d. Based on insertion time
-

38. Which of the following is true about domains?
- a. Two attributes cannot share the same domain
 - b. A domain cannot have a data format

- c. The same domain can be used for multiple attributes with different meanings
 - d. Domains are defined per tuple, not per attribute
-

39. If $R(A,B)$ has $\text{dom}(A) = \{1,2\}$ and $\text{dom}(B) = \{x\}$, what is the maximum number of tuples in any valid state $r(R)$?
- a. 1
 - b. 2
 - c. 3
 - d. 4
-

40. Which notation refers to a subtuple of t containing values of attributes A and C ?
- a. $t[A \cup C]$
 - b. $t[A, C]$
 - c. $t\{A, C\}$
 - d. $t.A.C$
-

41. What distinguishes a candidate key from a superkey?
- a. Candidate keys can contain nulls
 - b. Superkeys are always composite
 - c. Candidate keys are minimal
 - d. Superkeys cannot be primary keys
-

42. If a relation has two candidate keys, how many primary keys can it have?
- a. Zero
 - b. One
 - c. Two
 - d. As many as there are candidate keys
-

43. Which of the following update actions is **not** a standard option for handling referential integrity violation on DELETE?
- a. CASCADE
 - b. SET DEFAULT
 - c. SET NULL
 - d. RESTRICT

-
44. When modifying a non-key, non-foreign-key attribute, which constraint can be violated?
- a. Entity integrity
 - b. Referential integrity
 - c. Domain constraint
 - d. Key constraint
-

45. Which of the following is an inherent constraint of the relational model?
- a. Primary keys must be underlined
 - b. Tuples must be ordered
 - c. Attribute values must be atomic
 - d. Foreign keys must be indexed
-

46. In the relation state $r(R) = \{ \langle 0, a \rangle, \langle 0, b \rangle, \langle 1, c \rangle \}$, with $\text{dom}(A1) = \{0, 1\}$, $\text{dom}(A2) = \{a, b, c\}$, which of the following is **not** in the Cartesian product?
- a. $\langle 1, a \rangle$
 - b. $\langle 0, c \rangle$
 - c. $\langle 2, a \rangle$
 - d. $\langle 1, b \rangle$
-

47. What is the term for the current set of tuples in a relation?
- a. Schema
 - b. Extension
 - c. Intension
 - d. Domain
-

48. Which of the following best describes "entity integrity"?
- a. Every foreign key must reference a valid primary key
 - b. Primary key attributes must not be null
 - c. All attributes must have atomic values
 - d. Tuples must be unique
-

49. In a relational schema diagram, primary key attributes are typically:
- a. Bolded

- b. Italicized
 - c. Underlined
 - d. Colored red
-

50. Which of the following is **not** a valid reason for using a surrogate key?

- a. Natural key is too large
 - b. Natural key may change
 - c. Natural key contains nulls
 - d. To enforce referential integrity
-

51. If an INSERT operation provides a foreign key value that does not exist in the referenced relation, which constraint is violated?

- a. Entity integrity
 - b. Domain constraint
 - c. Referential integrity
 - d. Key constraint
-

52. Which of the following is a very hard distinction in relational theory?

- a. Schema vs. state
 - b. Tuple vs. attribute
 - c. Superkey vs. candidate key
 - d. Relation vs. table
-

53. The expression $t[A_u, A_v, \dots, A_w]$ denotes:

- a. A new relation
 - b. A subtuple of t
 - c. A domain intersection
 - d. A foreign key path
-

54. Which of the following is **not** part of the formal definition of a relation?

- a. Set of attributes
- b. Ordered tuples
- c. Allowance of duplicate rows
- d. Domains for each attribute

55. In the COMPANY database, if EMPLOYEE has Dno as a foreign key to DEPARTMENT.Dnumber, and DEPARTMENT is deleted, what happens under CASCADE DELETE?

- a. Only the department is deleted
 - b. The Dno in EMPLOYEE is set to NULL
 - c. All employees in that department are deleted
 - d. The delete is rejected
-

56. What is the main purpose of a relational database schema?

- a. Store current data
 - b. Define the logical structure and constraints
 - c. Optimize query execution
 - d. Manage user permissions
-

57. Which of the following constraints is checked **only** when a tuple is inserted or modified?

- a. Referential integrity on delete
 - b. Entity integrity
 - c. Domain constraint
 - d. Superkey uniqueness
-

58. If a relation schema includes an attribute Age derived from BirthDate, but the model stores only BirthDate, then Age is:

- a. A stored attribute
 - b. A composite attribute
 - c. Not part of the relational model (since derived attributes aren't stored)
 - d. A domain constraint
-

59. Which of the following is **not** a type of constraint classified in the chapter?

- a. Inherent
 - b. Schema-based
 - c. Application-based
 - d. Physical
-

60. The relational model's foundation is based on:

- a. Graph theory
 - b. Set theory
 - c. Predicate logic
 - d. Object-oriented principles
-

61. What is the minimum number of attributes a candidate key can have?

- a. 0
 - b. 1
 - c. 2
 - d. Depends on the relation
-

62. If a foreign key allows NULL, and it is **not** part of the primary key, what does a NULL typically mean?

- a. Error in data entry
 - b. The relationship is unknown or inapplicable
 - c. The referenced tuple was deleted
 - d. The attribute is composite
-

63. Which of the following is a very hard implication of the definition $r(R) \subset \text{dom}(A_1) \times \dots \times \text{dom}(A_n)$?

- a. Relations can have infinite tuples
 - b. Every tuple must conform to domain definitions
 - c. Attributes can have multiple domains
 - d. Tuples are ordered by domain size
-

64. In a self-describing tuple representation like $\{ \langle \text{name}, \text{"John"} \rangle, \langle \text{SSN}, 123 \rangle \}$, what is the advantage?

- a. Faster query processing
 - b. No need for schema
 - c. Attribute names are paired with values, removing ordering dependency
 - d. Supports multivalued attributes
-

65. Which update scenario is **most complex** in terms of constraint checking?

- a. Inserting a tuple with all non-key attributes

- b. Deleting a tuple not referenced by others
 - c. Updating a primary key that is widely referenced
 - d. Modifying a domain-constrained attribute to a valid value
-

66. What does the acronym DBMS stand for in the context of this chapter?

- a. Not discussed
 - b. Data Base Management Schema
 - c. Database Model System
 - d. Not applicable—focus is on relational model, not DBMS types
-

67. Which of the following is **not** mentioned as a way to handle constraint violations?

- a. Cancel the operation
 - b. Inform the user and proceed
 - c. Roll back the transaction
 - d. Execute a user-specified error-correction routine
-

68. The term "relation state" is synonymous with:

- a. Schema
 - b. Intension
 - c. Extension
 - d. Domain
-

69. If two relations R1 and R2 both have an attribute called "Name", but from different domains (e.g., PersonName vs. ProductName), this illustrates:

- a. Domain reuse with different roles
 - b. Schema redundancy
 - c. Referential integrity violation
 - d. Composite attribute usage
-

70. Which of the following best captures the essence of the relational model's mathematical basis?

- a. Tables with rows and columns
- b. Sets of n-tuples over attribute domains
- c. Graphs of entities and relationships
- d. Hierarchical data trees

Answer Key – Multiple Choice (1–70)

1. c
2. b
3. c
4. c
5. c
6. b
7. b
8. c
9. a
10. c
11. c
12. c
13. b
14. c
15. b
16. b
17. b
18. c
19. c
20. c
21. c
22. d
23. c
24. b
25. d
26. c
27. c
28. b
29. b
30. b
31. b
32. a
33. b
34. c
35. b

- 36. b
 - 37. b
 - 38. c
 - 39. b
 - 40. b
 - 41. c
 - 42. b
 - 43. b
 - 44. c
 - 45. c
 - 46. c
 - 47. b
 - 48. b
 - 49. c
 - 50. d
 - 51. c
 - 52. a
 - 53. b
 - 54. c
 - 55. c
 - 56. b
 - 57. c
 - 58. c
 - 59. d
 - 60. b
 - 61. b
 - 62. b
 - 63. b
 - 64. c
 - 65. c
 - 66. a (*Note: DBMS is not the focus of Chapter 5; the question tests attention to scope*)
 - 67. c (*Rollback is not explicitly listed; the text mentions cancel, inform, cascade, set null, user routine*)
 - 68. c
 - 69. a
 - 70. b
-

True/False Questions (30)

71. In the relational model, tuples are considered ordered.

- a. True
 - b. False
-

72. Attribute values in a tuple must be atomic.

- a. True
 - b. False
-

73. A superkey is always a candidate key.

- a. True
 - b. False
-

74. A candidate key is always a superkey.

- a. True
 - b. False
-

75. The primary key of a relation can contain null values if explicitly allowed.

- a. True
 - b. False
-

76. Referential integrity allows a foreign key to be null only if it is not part of the primary key of its relation.

- a. True
 - b. False
-

77. The relational model permits duplicate tuples in a relation state.

- a. True
 - b. False
-

78. The schema of a relation changes every time a tuple is inserted.

- a. True

b. False

79. Domain constraints are a type of explicit schema-based constraint.

- a. True
 - b. False
-

80. Semantic constraints can always be enforced using standard relational model constructs.

- a. True
 - b. False
-

81. The entity integrity constraint applies to all attributes in a relation.

- a. True
 - b. False
-

82. A relation state is a subset of the Cartesian product of its attribute domains.

- a. True
 - b. False
-

83. In the formal model, the order of attributes in a tuple is not significant.

- a. True
 - b. False
-

84. A surrogate key is a natural identifier like a social security number.

- a. True
 - b. False
-

85. Updating a foreign key can violate referential integrity.

- a. True
 - b. False
-

86. DELETE operations can violate entity integrity.

- a. True
 - b. False
-

87. INSERT operations can violate referential integrity.

- a. True
 - b. False
-

88. The COMPANY database schema includes integrity constraints as part of its definition.

- a. True
 - b. False
-

89. A null value is considered part of every domain.

- a. True
 - b. False
-

90. The relational model was proposed in 1970 by E.F. Codd.

- a. True
 - b. False
-

91. Every relation must have exactly one candidate key.

- a. True
 - b. False
-

92. If a relation has no candidate keys, it cannot exist in a valid relational database.

- a. True
 - b. False
-

93. The notation $t.A_i$ is equivalent to $t[A_i]$.

- a. True
- b. False

94. A foreign key must always reference the primary key of another relation.

- a. True
 - b. False
-

95. The CASCADE option on DELETE removes referencing tuples automatically.

- a. True
 - b. False
-

96. Inherent constraints include the atomicity of attribute values.

- a. True
 - b. False
-

97. A relational database state is also called a snapshot or instance.

- a. True
 - b. False
-

98. All attributes in a relation must have distinct domains.

- a. True
 - b. False
-

99. The primary key is used to provide tuple identity.

- a. True
 - b. False
-

100. The relational model allows multivalued attributes as long as they are enclosed in brackets.

- a. True
 - b. False
-
-

Answer Key – True/False (71–100)

- 71. b (False – tuples are unordered)
 - 72. a (True)
 - 73. b (False – superkey may not be minimal)
 - 74. a (True)
 - 75. b (False – primary key cannot be null)
 - 76. a (True)
 - 77. b (False – relations are sets, so no duplicates)
 - 78. b (False – schema is fixed; state changes)
 - 79. a (True – mentioned as another schema-based constraint)
 - 80. b (False – semantic constraints require triggers/assertions)
 - 81. b (False – only primary key attributes)
 - 82. a (True)
 - 83. b (False – in the formal definition used, attributes are ordered)
 - 84. b (False – surrogate is artificial)
 - 85. a (True)
 - 86. b (False – DELETE doesn't cause entity integrity violation)
 - 87. a (True)
 - 88. a (True – schema includes IC)
 - 89. b (False – null is special, not part of domain per se)
 - 90. a (True)
 - 91. b (False – can have multiple or, theoretically, none, though impractical)
 - 92. a (True – every relation must have a key to be valid)
 - 93. a (True)
 - 94. a (True – by definition in this context)
 - 95. a (True)
 - 96. a (True)
 - 97. a (True – called snapshot or instance)
 - 98. b (False – domains can be reused)
 - 99. a (True)
 - 100. b (False – multivalued attributes violate atomicity)
-